AI Assignment 3 - 24K0546

Student Information

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1. Competition Overview

- Competition Name: Predicting Irrigation Need
- Problem Type: Multiclass Classification
- Evaluation Metric: Balanced Accuracy Score
- Brief understanding of the dataset:

My target is irrigation_need with 3 classes **(Low, Medium, High)** so a multiclass classification would be needed with balanced accuracy (as per kaggle requirements), the dataset consists of 21 columns with the last one - target (irrigation_need), & rest as

Labels:

id,Soil_Type,Soil_pH,Soil_Moisture,Organic_Carbon,Electrical_Conductivity,Temperature_C,Humidity,**Rainfall_mm**,Sunlight_Hours,Wind_Speed_kmh,Crop_Type,Crop_Growth_Stage,Season,Irrigation_Type,Water_Source,Field_Area_hectare,Mulching_Used,**Previous_Irrigation_mm**,Region,

with Soil_type *(Clay, Loamy, Sandy, Silt)*, Crop_Type *(Cotton, Maze, Potato, Rice, Sugarcane, Wheat)*, Crop_Growth_Stage *(Flowering, Harvest, Sowing, Vegetative)*, Season (*Kharif, Rabi, Zaid)*, Irrigation_Type *(Canal, Drip, Rainfed, Sprinkler)*, Water_Source *(Groundwater, Rainwater, Reservoir, River)*, Mulching_Used *(Yes, No)*, and Region *(Central, North, East, West, South)* requiring encoding (as these are non-numerical)

2. Data Preprocessing

- Missing values handling: No missing values were found in the dataset, so no missing value handling was required.
- Encoding technique: The following required encoding as they had categorical data: Soil_Type, Crop_Type, Crop_Growth_Stage, Season, Irrigation_Type, Water_Source, Mulching_Used, Region. **Label Encoding** was used by assigning a *unique integer to each category.*
- Feature scaling: StandardScaler was applied to normalize all features to have mean of 0 & standard deviation of 1, this was necessary for distance based model (kNN).
- Train-validation split: data was divided into training & validation (with_held) set. 80% was set as training data & the other 20% for validation, making the **test_size 0.2**, with random_state=42 for reproducibility

3. Models Attempted

- KNN
- Naive Bayes
- Logistic Regression
- Decision Tree
- RandomForest (Advanced Model)

4. Cross Validation Strategy

- K-Fold details: k=5, All models were evaluated using 5-Fold cross validation on the data set. The data was divided into 5 equal chunks & the model was trained 5 times each time using a different chunk as the validation set. The 5 scores were averaged for the final test score.
- LOOCV observations: requires training data set for N times where N is the number of rows, the data set as 629999 rows which is computationally infeasible.
- Best validation accuracy: Decision Tree (max_depth=10) achieved the best CV score of 0.9608 ± 0.0009

5. Failed Attempts and Insights

- Model used - kNN

- Accuracy obtained

k=5: 0.4018

k=11: 0.3923

- Confusion matrix screenshot - What went wrong?

```
(.venv) ismailsilat@Ismails-MacBook-Air Assignment3 % python train.py train.csv
Model: KNN (k=5)
CV Score (5-Fold): 0.3993 ± 0.0015
Test Score:        0.4018
Confusion Matrix:
[[53788 19869    80]
 [32288 15034   692]
 [ 1883  1674   692]]
```

```
(.venv) ismailsilat@Ismails-MacBook-Air Assignment3 % python train.py train.csv
Model: KNN (k=11)
CV Score (5-Fold): 0.3831 ± 0.0020
Test Score:        0.3923
Confusion Matrix:
[[59014 14662    61]
 [36122 11285   607]
 [ 2142  1505   602]]
```

KNN performed poorly because the dataset is high-dimensional and contains both numerical and encoded categorical features. Distance-based methods struggle in such spaces, especially without feature scaling.

- What was improved?

Increasing the value of k slightly reduced noise but did not significantly improve performance. The model still remained unstable and biased toward majority classes.

- Model used - Naive Bayes

- Accuracy obtained :0.5615

- Confusion matrix screenshot - What went wrong?

```
(.venv) ismailsilat@Ismails-MacBook-Air Assignment3 % python train.py train.csv
Model: Naive Bayes
CV Score (5-Fold): 0.5587 ± 0.0018
Test Score:        0.5615
Confusion Matrix:
[[65496  8240     1]
 [20490 27313   211]
 [   92  3191   966]]
```

Naive Bayes assumes that all features are independent, which is not true in this dataset (e.g., rainfall, humidity, and soil moisture are related). This caused incorrect probability estimations.

- What was improved?

Despite its simplicity, Naive Bayes performed better than KNN, but still failed to capture complex relationships between features.

- Model used - Logistic Regression

- Accuracy obtained: 0.5381

- Confusion matrix screenshot - What went wrong?

```
(.venv) ismailsilat@Ismails-MacBook-Air Assignment3 % python train.py train.csv
Increase the number of iterations to improve the convergence (max_iter=5000).
You might also want to scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(

----------------------------------
Model: Logistic Regression
CV Score (5-Fold): 0.5494 ± 0.0078
Test Score:        0.5381
Confusion Matrix:
[[61174 12546    17]
 [18406 29241   367]
 [   90  3413   746]]
```

Logistic Regression struggled due to non-linear relationships between features and class labels. Additionally, convergence warnings indicated that the model did not fully optimize due to lack of feature scaling.

- What was improved?

Increasing maximum iterations improved stability slightly, but performance remained limited.

- Model used - Decision Tree

- Accuracy obtained

Default: 0.9447

Depth = 10: 0.9593 (better result)

Depth = 7 (tuned): **0.9619 (best result)**

- Confusion matrix screenshot - What went wrong?

```
(.venv) ismailsilat@Ismails-MacBook-Air Assignment3 % python train.py train.csv
Model: Decision Tree
CV Score (5-Fold): 0.9465 ± 0.0021
Test Score:        0.9447
Confusion Matrix:
[[72262  1475     0]
 [ 1260 46264   490]
 [    0   465  3784]]
```

```
(.venv) ismailsilat@Ismails-MacBook-Air Assignment3 % python train.py train.csv
Model: Decision Tree (d=10)
CV Score (5-Fold): 0.9608 ± 0.0009
Test Score:        0.9593
Confusion Matrix:
[[73413   323     1]
 [ 1025 46823   166]
 [    0   395  3854]]
```

```
(.venv) ismailsilat@Ismails-MacBook-Air Assignment3 % python train.py train.csv
Model: Decision Tree (d=7)
CV Score (5-Fold): 0.9619 ± 0.0013
Test Score:        0.9589
Confusion Matrix:
[[73429   308     0]
 [ 1006 46786   222]
 [    0   397  3852]]
```

Default decision trees tend to overfit when depth is unrestricted.

- What was improved?

Hyperparameter tuning via RandomizedSearchCV on additional parameters (min_samples_split, min_samples_leaf, criterion, max_features, min_impurity_decrease) with depth fixed at 7 yielded the best overall result.

- Model used - Random Forest

- Accuracy obtained: 0.9562

- Confusion matrix screenshot - What went wrong?

```
(.venv) ismailsilat@Ismails-MacBook-Air Assignment3 % python train.py train.csv
Model: Random Forest
CV Score (5-Fold): 0.9574 ± 0.0010
Test Score:        0.9562
Confusion Matrix:
[[73423   314     0]
 [ 1004 46908   102]
 [    0   442  3807]]
```

Although Random Forest performed strongly, it did not outperform the tuned Decision Tree in this case.

- What was improved?

Ensemble learning improved stability compared to a single tree, reducing variance in predictions. **Further hyperparameter tuning (e.g. n_estimators, max_depth, max_features) was not pursued due to the high computational cost of training multiple trees on 630,000 rows**, which made exhaustive search impractical within time constraints.

6. Final Model Selection

Best model: **Decision Tree Classifier (max_depth = 10)**

- Hyperparameters

max_depth = 7

min_samples_split = 5

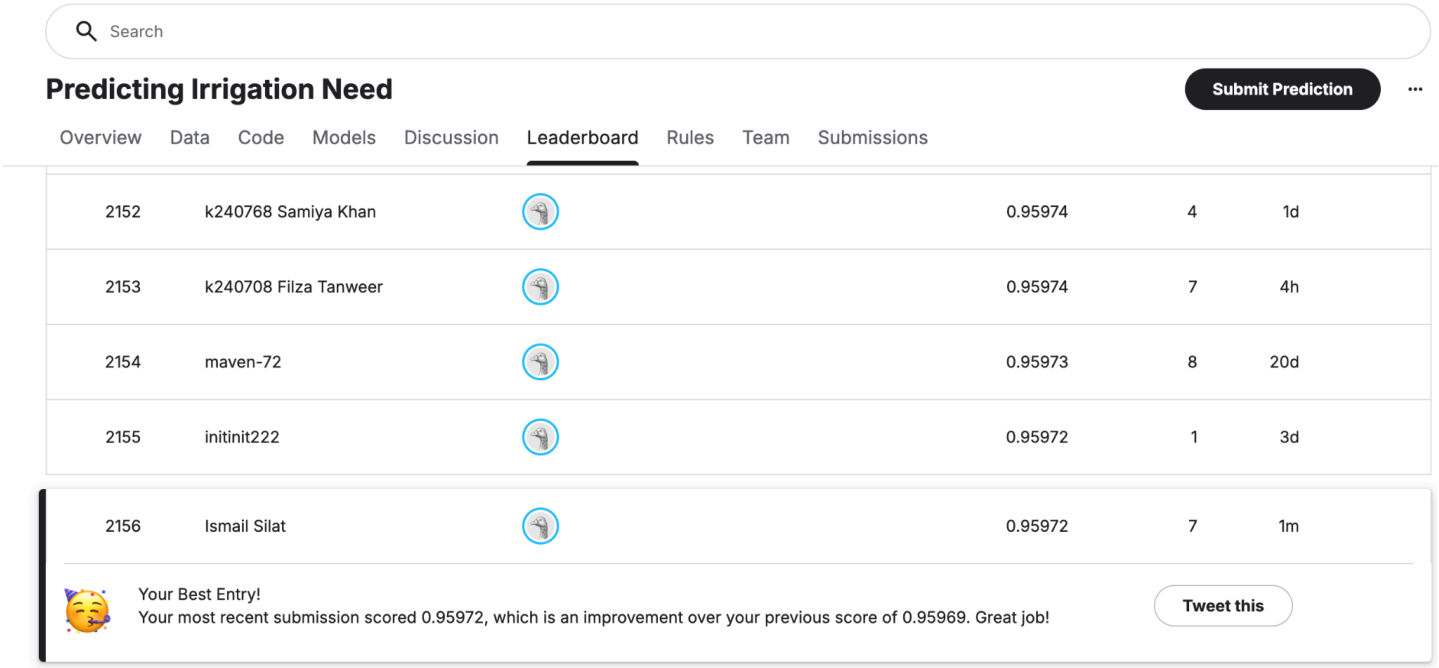
min_samples_leaf = 8
min_impurity_decrease = 0.0
max_features = None
criterion = entropy
random_state = 42

- Why selected?

This model was selected because it achieved the **highest balanced accuracy score (0.9619)** among all tested models. It also showed; **strong performance** across all three classes (Low, Medium, High), reduced overfitting compared to an unrestricted decision tree, and **better generalization** than Logistic Regression, Naive Bayes, and KNN. Compared to Random Forest, it performed slightly better while being simpler and easier to interpret with **lower computational costs**. Hyperparameters were found using RandomizedSearchCV over 20 random combinations, searching on a 30% stratified sample of training data for efficiency.

7. Leaderboard Performance - Kaggle score

- Rank **2738**
- Screenshot attached



Submission with name: Ismail Silat

8. Conclusion and Learnings

- Key insights

This project demonstrated that tree-based models are highly effective for multiclass classification problems involving mixed numerical and categorical features. Decision Trees and Random Forests outperformed linear and distance-based models.

- Challenges faced

1. Encoding multiple categorical variables correctly
2. Ensuring fair comparison across models
3. Handling convergence issues in Logistic Regression
4. Interpreting confusion matrices for multiclass outputs

- Future improvements

1. Applying feature scaling to improve Logistic Regression performance
2. Using advanced ensemble models
3. Performing hyperparameter tuning
4. Exploring feature selection to reduce noise and improve generalization